

REDIS

1. Install redis without internet

1.1. Download Packed Redis

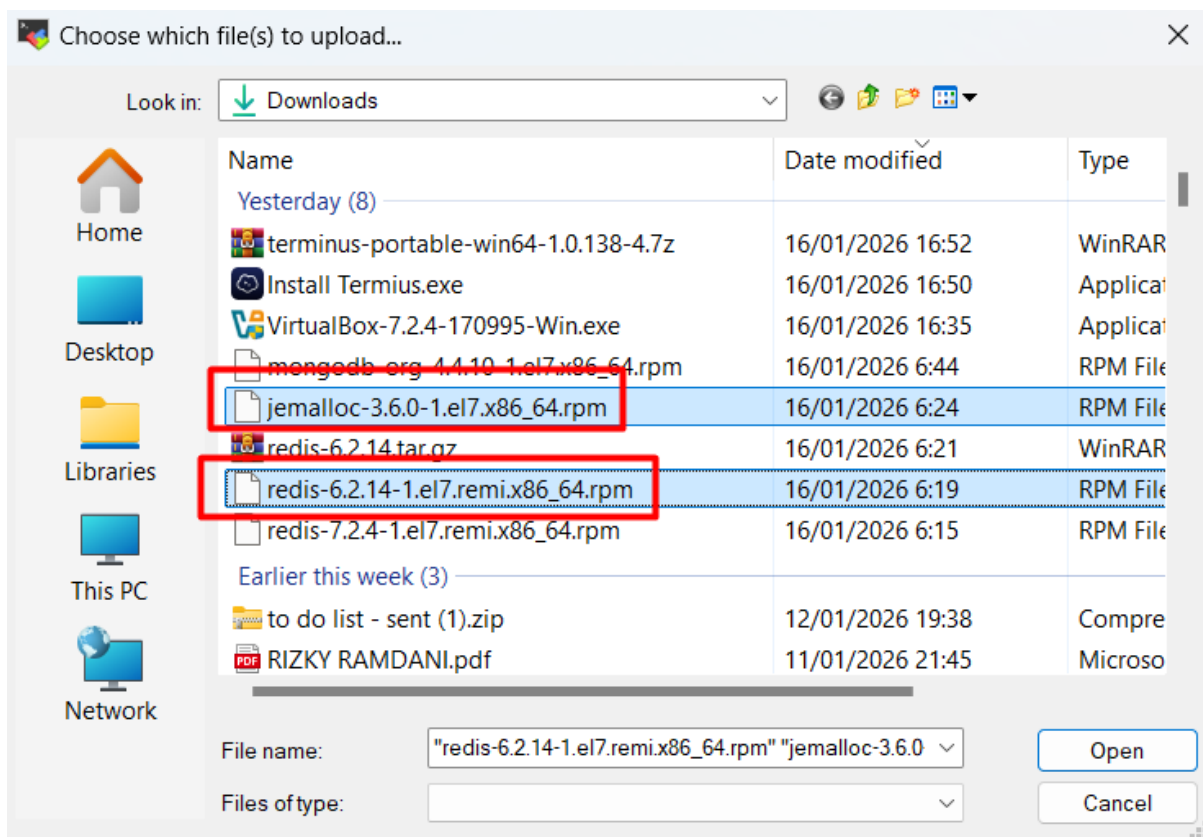
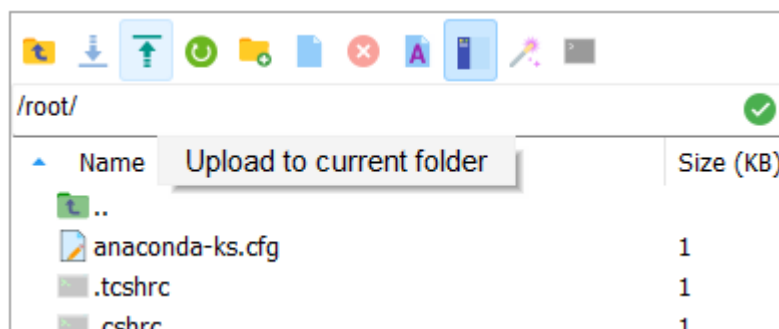
Download RPM Redis

<https://rpmfind.net/linux/rpm2html/search.php?query=redis&submit=Search>

Download Dependency jemalloc redis

<https://rpmfind.net/linux/rpm2html/search.php?query=jemalloc>

1.2. Upload File



/root/	
Name	Size (KB)
redis-6.2.14-1.el7.remix86_64.rpm	1 202
jemalloc-3.6.0-1.el7.x86_64.rpm	104
anaconda-ks.cfg	1
tesha	1

```
[root@belajar ~]# ls -l
total 1316
-rw-----. 1 root root 1274 Jan 16 16:51 anaconda-ks.cfg
-rw-r--r--. 1 root root 107420 Jan 16 17:02 jemalloc-3.6.0-1.el7.x86_64.rpm
-rw-r--r--. 1 root root 1231020 Jan 16 17:02 redis-6.2.14-1.el7.remix86_64.rpm
[root@belajar ~]#
```

1.3. Install Redis

Install Dependency jemalloc redis terlebih dahulu

```
rpm -ivh jemalloc-3.6.0-1.el7.x86_64.rpm
```

```
[root@belajar ~]# rpm -ivh jemalloc-3.6.0-1.el7.x86_64.rpm
warning: jemalloc-3.6.0-1.el7.x86_64.rpm: Header V4 RSA/SHA1 Signature, key ID 764429e6: NOKEY
Preparing...
Updating / installing...
 1:jemalloc-3.6.0-1.el7
[root@belajar ~]#
```

Setelah berhasil Install packed Redis

```
rpm -ivh redis-6.2.14-1.el7.remix86_64.rpm
```

```
[root@belajar ~]# rpm -ivh redis-6.2.14-1.el7.remix86_64.rpm
warning: redis-6.2.14-1.el7.remix86_64.rpm: Header V4 DSA/SHA1 Signature, key ID 00f97f56: NOKEY
Preparing...
Updating / installing...
 1:redis-6.2.14-1.el7.remix86_64
[root@belajar ~]#
```

1.4. Start dan Enable Service

```
systemctl status redis
systemctl start redis
systemctl enable redis
systemctl status redis
```

```
[root@belajar ~]# systemctl status redis
● redis.service - Redis persistent key-value database
   Loaded: loaded (/usr/lib/systemd/system/redis.service; disabled; vendor preset: disabled)
   Drop-In: /etc/systemd/system/redis.service.d
            └─limit.conf
   Active: inactive (dead)
[root@belajar ~]#
[root@belajar ~]# systemctl start redis
[root@belajar ~]# systemctl status redis
● redis.service - Redis persistent key-value database
   Loaded: loaded (/usr/lib/systemd/system/redis.service; disabled; vendor preset: disabled)
   Drop-In: /etc/systemd/system/redis.service.d
            └─limit.conf
   Active: active (running) since Sat 2026-01-17 05:43:42 WIB; 1s ago
 Main PID: 2964 (redis-server)
   Status: "Ready to accept connections"
   CGroup: /system.slice/redis.service
           └─2964 /usr/bin/redis-server 127.0.0.1:6379

Jan 17 05:43:42 belajar.localdomain systemd[1]: Starting Redis persistent key-value database...
Jan 17 05:43:42 belajar.localdomain systemd[1]: Started Redis persistent key-value database.
[root@belajar ~]#
[root@belajar ~]#
[root@belajar ~]# systemctl enable redis
Created symlink from /etc/systemd/system/multi-user.target.wants/redis.service to /usr/lib/systemd/system/redis.service.
[root@belajar ~]# systemctl status redis
● redis.service - Redis persistent key-value database
   Loaded: loaded (/usr/lib/systemd/system/redis.service; enabled; vendor preset: disabled)
   Drop-In: /etc/systemd/system/redis.service.d
            └─limit.conf
   Active: active (running) since Sat 2026-01-17 05:43:42 WIB; 13s ago
 Main PID: 2964 (redis-server)
   Status: "Ready to accept connections"
   CGroup: /system.slice/redis.service
           └─2964 /usr/bin/redis-server 127.0.0.1:6379

Jan 17 05:43:42 belajar.localdomain systemd[1]: Starting Redis persistent key-value database...
Jan 17 05:43:42 belajar.localdomain systemd[1]: Started Redis persistent key-value database.
[root@belajar ~]#
```

```
[root@belajar ~]# redis-server --version
Redis server v=6.2.14 sha=00000000:0 malloc=jemalloc-5.1.0 bits=64 build=433b4d80795b2a8f
[root@belajar ~]#
```

1.5. Testing Redis

```
redis-cli ping
```

```
[root@belajar ~]# redis-cli ping
PONG
[root@belajar ~]#
```

1.6. Set Password Redis

```
vi /etc/redis.conf
-- cari # requirepass foobared lalu ubah
contoh : requirepass welcome123
```

```
[root@belajar ~]# vi /etc/redis.conf
[root@belajar ~]#
```

```
[root@belajar ~]# cat /etc/redis.conf | grep welcome123
requirepass welcome123
[root@belajar ~]#
```

1.7. Test Login ke Redis

```
[root@belajar ~]# redis-cli AUTH welcome123
OK
[root@belajar ~]#
```

Jika belum login maka akan error

```
[root@belajar ~]# redis-cli
127.0.0.1:6379> select 1
(error) NOAUTH Authentication required.
127.0.0.1:6379> AUTH welcome123
OK
127.0.0.1:6379> select 1
OK
127.0.0.1:6379[1]> exit
[root@belajar ~]#
```

2. Administrasi Redis

Create Database

Jika di RDBMS seperti postgresql, mysql, sql dll itu create database. Di redis tidak langsung masuk ke databasenya dengan command **select 1**. Maka database sudah dibuat atau sudah masuk ke database 1.

```
[root@belajar ~]# redis-cli
127.0.0.1:6379> AUTH welcome123
OK
127.0.0.1:6379> select 1
OK
127.0.0.1:6379[1]>
```

DML

Create table dan insert data di Redis

Berikut konsepnya;

RDBMS	REDIS
Database └─ Schema └─ Table users └─ Row id=1 └─ Row id=2	Database 1 └─ users:1 (HASH) └─ users:2 (HASH) └─ site:name (STRING)

RDBMS Table	Redis Table
TABLE users └─ ROW id=1 └─ ROW id=2 └─ ROW id=3	users:1 ← 1 ROW users:2 ← 1 ROW users:3 ← 1 ROW
JIKA RDBMS buat table dahulu users (id INT, name VARCHAR, age INT, email VARCHAR)	Redis langsung insert HSET users:1 id 1 name "Andi" age 30 email "andi@mail.com" HSET users:2 id 2 name "Ani" age 25 email "ani@mail.com"

CREATE TABLE DAN INSERT

Lansung insert data pada redis tidak perlu buat table

```
127.0.0.1:6379[1]> HSET users:1 id 1 name "Andi" age 30 email "andi@mail.com"
(integer) 4
127.0.0.1:6379[1]> HSET users:2 id 2 name "Ani" age 25 email "ani@mail.com"
(integer) 4
127.0.0.1:6379[1]> HSET users:3 id 3 name "Budi" age 28 email "budi@mail.com"
(integer) 4
127.0.0.1:6379[1]> HSET users:4 id 4 name "Citra" age 32 email "citra@mail.com"
(integer) 4
127.0.0.1:6379[1]> HSET users:5 id 5 name "Dedi" age 27 email "dedi@mail.com"
(integer) 4
127.0.0.1:6379[1]> HSET users:6 id 6 name "Eka" age 35 email "eka@mail.com"
(integer) 4
127.0.0.1:6379[1]> HSET users:7 id 7 name "Fajar" age 29 email "fajar@mail.com"
(integer) 4
127.0.0.1:6379[1]> HSET users:8 id 8 name "Gina" age 26 email "gina@mail.com"
(integer) 4
127.0.0.1:6379[1]> HSET users:9 id 9 name "Hadi" age 31 email "hadi@mail.com"
(integer) 4
127.0.0.1:6379[1]> HSET users:10 id 10 name "Intan" age 24 email "intan@mail.com"
(integer) 4
127.0.0.1:6379[1]>
```

SELECT DATA

Jika di RDBMS select * from nama table. Untuk di redis hanya menggunakan SCAN 0 MATCH users:*

```
127.0.0.1:6379[1]> SCAN 0 MATCH users:*
1) "0"
2) 1) "users:3"
   2) "users:2"
   3) "users:1"
   4) "users:6"
   5) "users:5"
   6) "users:8"
   7) "users:7"
   8) "users:10"
   9) "users:4"
  10) "users:9"
127.0.0.1:6379[1]>
```

Jika ingin SELECT * FROM users WHERE id = 7 di RDBMS. Maka untuk select berdasarkan WHERE di Redis menggunakan HGETALL users:7

```
127.0.0.1:6379[1]> HGETALL users:7
1) "id"
2) "7"
3) "name"
4) "Fajar"
5) "age"
6) "29"
7) "email"
8) "fajar@mail.com"
127.0.0.1:6379[1]>
```

Jika ingin SELECT name, age FROM users WHERE id = 3 di RDBMS. Maka di Redis HMGET users:3 name age

```
127.0.0.1:6379[1]> HMGET users:3 name age
1) "Budi"
2) "28"
127.0.0.1:6379[1]>
```

UPDATE

Jika update data. Untuk command sql di RDBMS UPDATE users SET age=40 WHERE id=6;

Maka untuk di Redis cukup command HSET users:6 age 40

Before

```
127.0.0.1:6379[1]> HGETALL users:6
1) "id"
2) "6"
3) "name"
4) "Eka"
5) "age"
6) "35"
7) "email"
8) "eka@mail.com"
127.0.0.1:6379[1]>
```

After

```
127.0.0.1:6379[1]> HSET users:6 age 40
(integer) 0
127.0.0.1:6379[1]> HGETALL users:6
1) "id"
2) "6"
3) "name"
4) "Eka"
5) "age"
6) "40"
7) "email"
8) "eka@mail.com"
127.0.0.1:6379[1]>
```

DELETE DATA

DELETE FROM user WHERE id = 4 jika di database RDBMS. Untuk Redis sendiri menggunakan command DEL users:4

Drop database di Redis

```
127.0.0.1:6379> select 1
OK
127.0.0.1:6379[1]> FLUSHDB
OK
127.0.0.1:6379[1]> KEYS *
(empty array)
127.0.0.1:6379[1]>
```

3. Backup Database Redis

3.1. Backup RDB (Redis Database Snapshot)

Backup snapshot full database dalam satu file dump.rdp

Cara kerjanya database Redis menyimpan seluruh data di **memory** lalu transfer ke disk pada waktu tertentu. Berikut cara kerj backup-nya; (jika pakai password untuk login redis pastikan login dengan password)

1. SAVE (Blocking (Redis berhenti sebentar). Cocok untuk maintenance)

```
[root@belajar ~]# redis-cli -a welcome123
Warning: Using a password with '-a' or '-u' option on the command line interface may not be safe.
127.0.0.1:6379> SAVE
OK
127.0.0.1:6379> exit
[root@belajar ~]#
```

Lokasi default tempat penyimpanan

```
[root@belajar ~]# cd /var/lib/redis/
[root@belajar redis]# ls -altrh
total 8.0K
drwxr-xr-x. 26 root root 4.0K Jan 17 05:43 ..
-rw-r--r--. 1 redis redis 720 Jan 17 11:49 dump.rdb
drwxr-x---. 2 redis redis 22 Jan 17 11:49 .
[root@belajar redis]#
```

2. BGSAVE (Redis tetap melayani request, backup berjalan di background)

```
[root@belajar redis]# redis-cli
127.0.0.1:6379> AUTH welcome123
OK
127.0.0.1:6379> BGSAVE
Background saving started
127.0.0.1:6379>
```

Lokasi default tempat penyimpanan

```
[root@belajar ~]# cd /var/lib/redis/
[root@belajar redis]# ls -altrh
total 12K
drwxr-xr-x. 26 root root 4.0K Jan 17 05:43 ..
-rw-r--r--. 1 redis redis 720 Jan 17 11:49 dump_save.rdb
-rw-r--r--. 1 redis redis 720 Jan 17 12:03 dump.rdb
drwxr-x---. 2 redis redis 43 Jan 17 12:03 .
[root@belajar redis]#
```

Kelebihan	Kekurangan
File kecil & cepat restore	Data loss sejak snapshot terakhir
Cocok untuk backup harian	Tidak real-time
Overhead rendah	Tidak point-in-time recovery
Mudah dipindahkan (scp/rsync)	

3.2. Backup AOF (Append Only File)

Mencatat setiap perintah write (SET, DEL, INCR, dll)

Untuk menggunakan metode ini aktifkan dulu **Append Only**. Ada 2 cara aktifkannya;

1. redis-cli -a PASSWORD_ANDA CONFIG SET appendonly yes hanya aktif sementara dan tanpa perlu restart service. (Tidak permanen)
2. Langsung dari confi nya vi /etc/redis.conf cari appendonly no lalu ubah jadi **appendonly yes** setelah diubah restart service. (Permanen)

Disini pakai cara 2 untuk aktifkan AOF nya.

Sebelum

```
[root@belajar ~]# cat /etc/redis.conf | grep appendonly
appendonly no
# The name of the append only file (default: "appendonly.aof")
appendfilename "appendonly.aof"
[root@belajar ~]#
```

vi /etc/redis.conf (ubah **appendonly no** → **appendonly yes**)
systemctl status redis
systemctl restart redis
systemctl status redis

Sesudah

```
[root@belajar ~]# cat /etc/redis.conf | grep appendonly
appendonly yes
# The name of the append only file (default: "appendonly.aof")
appendfilename "appendonly.aof"
[root@belajar ~]#
```

Cara backup AOF. Pastikan aof_enabled:1

```
[root@belajar ~]# redis-cli
127.0.0.1:6379> AUTH welcome123
OK
127.0.0.1:6379> INFO persistence
# Persistence
loading:0
current_cow_size:0
current_cow_size_age:0
current_fork_perc:0.00
current_save_keys_processed:0
current_save_keys_total:0
rdb_changes_since_last_save:0
rdb_bgsave_in_progress:0
rdb_last_save_time:1768628697
rdb_last_bgsave_status:ok
rdb_last_bgsave_time_sec:-1
rdb_current_bgsave_time_sec:-1
rdb_last_cow_size:0
aof_enabled:1
aof_rewrite_in_progress:0
aof_rewrite_scheduled:0
aof_last_rewrite_time_sec:-1
aof_current_rewrite_time_sec:-1
aof_last_bgrewrite_status:ok
aof_last_write_status:ok
aof_last_cow_size:0
module_fork_in_progress:0
module_fork_last_cow_size:0
aof_current_size:720
aof_base_size:720
aof_pending_rewrite:0
aof_buffer_length:0
aof_rewrite_buffer_length:0
aof_pending_bio_fsync:0
aof_delayed_fsync:0
127.0.0.1:6379>
```

Selanjutnya jalankan backup AOF nya

BGREWRITEAOF

```
127.0.0.1:6379> BGREWRITEAOF
Background append only file rewriting started
127.0.0.1:6379>
```

Cek file backup AOF

```
[root@belajar ~]# ls -altrh /var/lib/redis/
total 16K
drwxr-xr-x. 26 root root 4.0K Jan 17 05:43 ..
-rw-r--r--. 1 redis redis 720 Jan 17 11:49 dump_save.rdb
-rw-r--r--. 1 redis redis 720 Jan 17 12:44 dump.rdb
-rw-r--r--. 1 redis redis 720 Jan 17 12:58 appendonly.aof
drwxr-x---. 2 redis redis 65 Jan 17 12:58 .
[root@belajar ~]#
```

Kelebihan	Kekurangan
Data loss minimal (≤ 1 detik)	File besar
Bisa replay data	Restore lebih lambat
Lebih aman dari RDB	Perlu rewrite (BGREWRITEAOF)

3.3. Backup RDB + AOF (Hybrid / Mixed Mode)

Menggabungkan snapshot **RDB + log AOF**.

RDB → Snapshot FULL atau Backup Full

AOF → Log perubahan (Backup incremental)

Langkah Backup = Load RDB dahulu → kemudian Lanjutkan AOF

1. Ubah konfigurasi redis.conf seperti gambar dibawah ini;

```
#
save 900 1
save 300 100
save 60 10000
```

```
auto-aof-rewrite-percentage 100
auto-aof-rewrite-min-size 64mb

# An AOF file may be found to be tr
```

```
[root@belajar ~]# cat /etc/redis.conf | grep append
appendonly yes
# The name of the append only file (default: "appendonly.a
appendfilename "appendonly.aof"
# always: fsync after every write to the append only log.
# appendfsync always
appendfsync everysec
# appendfsync no
# the same as "appendfsync none". In practical terms, this
no-appendfsync-on-rewrite no
# Automatic rewrite of the append only file.
# appending new stream entries. If any of the following se
```

Kemudian SAVE dan RESTART SERVICE REDIS

```
[root@belajar ~]# systemctl restart redis
[root@belajar ~]#
[root@belajar ~]# systemctl status redis
● redis.service - Redis persistent key-value database
   Loaded: loaded (/usr/lib/systemd/system/redis.service; enabled; vendor preset: disabled)
   Drop-In: /etc/systemd/system/redis.service.d
            └─limit.conf
   Active: active (running) since Sat 2026-01-17 13:24:32 WIB; 2s ago
   Process: 20113 ExecStop=/usr/libexec/redis-shutdown (code=exited, status=0/SUCCESS)
   Main PID: 20128 (redis-server)
   Status: "Ready to accept connections"
   CGroup: /system.slice/redis.service
           └─20128 /usr/bin/redis-server 127.0.0.1:6379

Jan 17 13:24:32 belajar.localdomain systemd[1]: Stopped Redis persistent key-value database.
Jan 17 13:24:32 belajar.localdomain systemd[1]: Starting Redis persistent key-value database..
Jan 17 13:24:32 belajar.localdomain systemd[1]: Started Redis persistent key-value database.
[root@belajar ~]#
```

2. Selanjutnya uji coba jalankan backup RDB + AOF
BGSAVE = Ini adalah backup full nya redis

```
[root@belajar ~]# redis-cli
127.0.0.1:6379> AUTH welcome123
OK
127.0.0.1:6379> BGSAVE
Background saving started
127.0.0.1:6379> █
```

```
[root@belajar ~]# cd /var/lib/redis/
[root@belajar redis]# ls -altrh
total 8.0K
drwxr-xr-x. 26 root root 4.0K Jan 17 05:43 ..
-rw-r--r--. 1 redis redis 720 Jan 17 14:36 dump.rdb
drwxr-x---. 2 redis redis 22 Jan 17 14:36 .
[root@belajar redis]#
```

BGREWRITEAOF = Ini adalah backup incrementalnya

```
127.0.0.1:6379> BGREWRITEAOF
Background append only file rewriting started
127.0.0.1:6379> █
```

```
[root@belajar redis]# ls -altrh
total 12K
drwxr-xr-x. 26 root root 4.0K Jan 17 05:43 ..
-rw-r--r--. 1 redis redis 720 Jan 17 14:36 dump.rdb
-rw-r--r--. 1 redis redis 720 Jan 17 14:37 appendonly.aof
drwxr-x---. 2 redis redis 44 Jan 17 14:37 .
[root@belajar redis]#
```



Kelebihan	Kekurangan
Restore cepat	File besar
Data loss minimal	Lebih banyak disk usage
Stabil di production	
Dipakai oleh Redis Enterprise	

Setting-an untuk level production yang di rekomendasikan. Kembali lagi tergantung performance hardware.

```
appendonly yes
appendfsync everysec

save 900 1
save 300 10
save 60 10000

auto-aof-rewrite-percentage 100
auto-aof-rewrite-min-size 64mb
```

3.4. Backup via Redis CLI Export

Backup ini seperti expdp di oracle jadi hanya backup table data tertentu.

Jadi Redis CLI Export mengambil data langsung dari redis via perintah CLI. Mengeksport key dan value ke file teks/script.

Berikut cara kerjanya;

Buat command dan pastikan ke database yang dituju. Contoh DB 1

```
redis-cli -a welcome123 -n 1 --scan | while read key; do
echo "SET \"$key\" \"$(redis-cli -a welcome123 -n 1 GET \"$key\")\""
done > redis_db1_export.txt
```

n 1 adalah database yang dituju dan untuk menarik semua key / data yang ada di database 1.

Sebelumnya dicek dulu database 1 ada apa aja didalamnya.

```
[root@belajar ~]# redis-cli
127.0.0.1:6379> AUTH welcome123
OK
127.0.0.1:6379> select 1
OK
```

```
127.0.0.1:6379[1]> SCAN 0 MATCH users:*
1) "0"
2) 1) "users:2"
   2) "users:3"
   3) "users:5"
   4) "users:10"
   5) "users:7"
   6) "users:6"
   7) "users:1"
   8) "users:9"
   9) "users:8"
127.0.0.1:6379[1]>
```

Contoh diatas database 1 ada key / data user isinya. Contoh buka key user 7

```
127.0.0.1:6379[1]> HGETALL users:7
1) "id"
2) "7"
3) "name"
4) "Fajar"
5) "age"
6) "29"
7) "email"
8) "fajar@mail.com"
127.0.0.1:6379[1]>
```

Berikutnya jalankan skrip yang sudah dibuat tadi

[illegible]

*Noted = untuk warning tersebut normal hanya pemberitahuan untuk perintah -a atau -u sangat tidak baik dijalankan di command. Jadi alangkah baiknya password tersebut disimpan nanti tinggal panggil, jadi seperti itu sarannya.

```
[root@belajar ~]# ls -l
total 8
-rw-r--r--. 1 root root 1295 Jan 17 14:34 anaconda-ks.cfg
-rw-r--r--. 1 root root 739 Jan 17 15:19 redis_db1_export.txt
[root@belajar ~]#
```



Setelah backup CLI Export maka akan terbuat skrip yang digunakan untuk restore.

```
[root@belajar ~]# cat redis_db1_export.txt
SET "users:2" "WRONGTYPE Operation against a key holding the wrong kind of value"
SET "users:3" "WRONGTYPE Operation against a key holding the wrong kind of value"
SET "users:5" "WRONGTYPE Operation against a key holding the wrong kind of value"
SET "users:10" "WRONGTYPE Operation against a key holding the wrong kind of value"
SET "users:7" "WRONGTYPE Operation against a key holding the wrong kind of value"
SET "users:6" "WRONGTYPE Operation against a key holding the wrong kind of value"
SET "users:1" "WRONGTYPE Operation against a key holding the wrong kind of value"
SET "users:9" "WRONGTYPE Operation against a key holding the wrong kind of value"
SET "users:8" "WRONGTYPE Operation against a key holding the wrong kind of value"
[root@belajar ~]#
```

Ketika dibuka file tersebut isinya perintah untuk insert.

Berikut cara agar tidak pakai password;

Pertama jalankan perintah ini = **export REDISCLI_AUTH='PASSWORD_ANDA'**

Kemudian jalankan skripnya tanpa -a atau -u

```
redis-cli -n 1 --scan | while read key; do
  echo "SET \"$key\" \"$(redis-cli -n 1 GET \"$key\")\""
done > redis_db1_export_cara2.txt
```

```
[root@belajar ~]# export REDISCLI_AUTH='welcome123'
[root@belajar ~]# redis-cli -n 1 --scan | while read key; do
>   echo "SET \"$key\" \"$(redis-cli -n 1 GET \"$key\")\""
> done > redis_db1_export_cara2.txt
[root@belajar ~]#
```

Selanjutnya cek apakah sudah berhasil atau belum untuk backup export-nya

```
[root@belajar ~]# ls -l
total 8
-rw-r--r--. 1 root root 1295 Jan 17 14:34 anaconda-ks.cfg
-rw-r--r--. 1 root root 739 Jan 17 15:41 redis_db1_export_cara2.txt
[root@belajar ~]#
[root@belajar ~]#
[root@belajar ~]# cat redis_db1_export_cara2.txt
SET "users:2" "WRONGTYPE Operation against a key holding the wrong kind of value"
SET "users:3" "WRONGTYPE Operation against a key holding the wrong kind of value"
SET "users:5" "WRONGTYPE Operation against a key holding the wrong kind of value"
SET "users:10" "WRONGTYPE Operation against a key holding the wrong kind of value"
SET "users:7" "WRONGTYPE Operation against a key holding the wrong kind of value"
SET "users:6" "WRONGTYPE Operation against a key holding the wrong kind of value"
SET "users:1" "WRONGTYPE Operation against a key holding the wrong kind of value"
SET "users:9" "WRONGTYPE Operation against a key holding the wrong kind of value"
SET "users:8" "WRONGTYPE Operation against a key holding the wrong kind of value"
[root@belajar ~]#
```

